

# ENTITY RELATIONSHIP (ER) DIAGRAM AND RELATIONAL SCHEMA

## Airline Reservation / Booking System

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### Objective

Design a complete Entity Relationship (ER) Diagram for the selected application/domain and convert it into a Relational Schema. The design identifies entities, attributes, primary keys, relationships, cardinalities, and foreign keys.

### 1. Entities and Attributes

#### PASSENGER

| Attribute    | Key | Description |
|--------------|-----|-------------|
| Passenger_ID | PK  | Primary Key |
| Name         |     | Attribute   |
| Email        |     | Attribute   |
| Passport_No  |     | Attribute   |
| Phone        |     | Attribute   |

#### BOOKING

| Attribute      | Key | Description |
|----------------|-----|-------------|
| Booking_ID     | PK  | Primary Key |
| Booking_Date   |     | Attribute   |
| Booking_Status |     | Attribute   |
| Passenger_ID   | FK  | Foreign Key |
| Flight_ID      | FK  | Foreign Key |
| Seat_ID        | FK  | Foreign Key |

#### PAYMENT

| Attribute      | Key | Description |
|----------------|-----|-------------|
| Payment_ID     | PK  | Primary Key |
| Amount         |     | Attribute   |
| Payment_Date   |     | Attribute   |
| Payment_Method |     | Attribute   |
| Payment_Status |     | Attribute   |
| Booking_ID     | FK  | Foreign Key |

## FLIGHT

| Attribute            | Key | Description |
|----------------------|-----|-------------|
| Flight_ID            | PK  | Primary Key |
| Flight_Number        |     | Attribute   |
| Departure_Time       |     | Attribute   |
| Arrival_Time         |     | Attribute   |
| Status               |     | Attribute   |
| Aircraft_ID          | FK  | Foreign Key |
| Departure_Airport_ID | FK  | Foreign Key |
| Arrival_Airport_ID   | FK  | Foreign Key |

## AIRPORT

| Attribute    | Key | Description |
|--------------|-----|-------------|
| Airport_ID   | PK  | Primary Key |
| Airport_Name |     | Attribute   |
| City         |     | Attribute   |
| Country      |     | Attribute   |

## AIRCRAFT

| Attribute    | Key | Description |
|--------------|-----|-------------|
| Aircraft_ID  | PK  | Primary Key |
| Model        |     | Attribute   |
| Capacity     |     | Attribute   |
| Manufacturer |     | Attribute   |

## SEAT

| Attribute   | Key | Description |
|-------------|-----|-------------|
| Seat_ID     | PK  | Primary Key |
| Seat_Number |     | Attribute   |
| Class       |     | Attribute   |
| Aircraft_ID | FK  | Foreign Key |

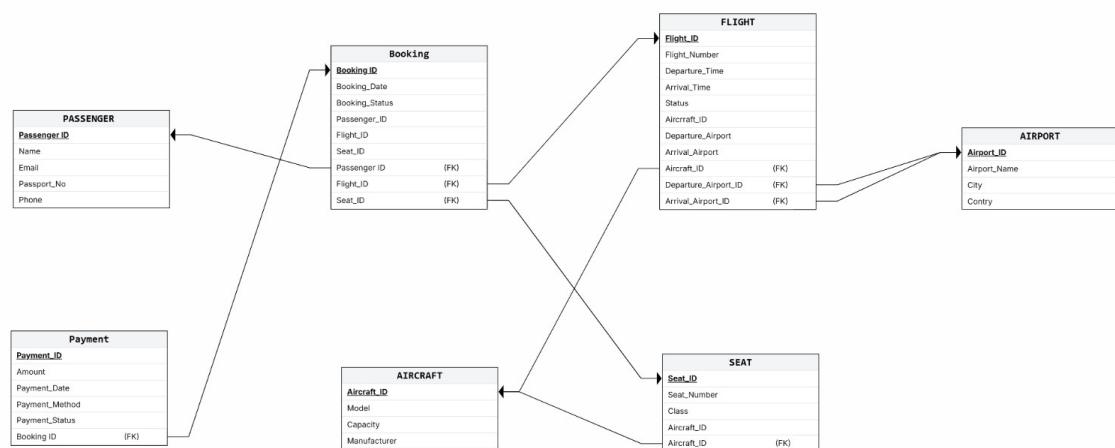
## 2. Relationships and Cardinalities

| Relationship | Entities            | Cardinality | Explanation                                                                  |
|--------------|---------------------|-------------|------------------------------------------------------------------------------|
| Makes        | PASSENGER — BOOKING | 1 : M       | One passenger can make many bookings; each booking belongs to one passenger. |
| Generates    | BOOKING — PAYMENT   | 1 : 1       | A booking generates a payment record in the designed model.                  |

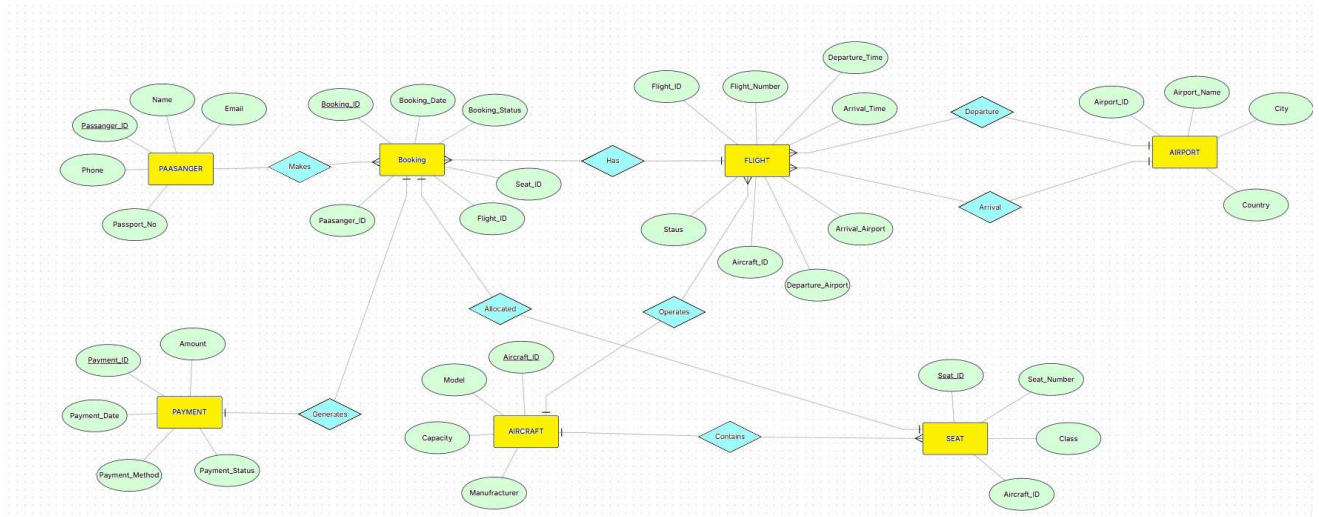
|           |                   |       |                                                                                  |
|-----------|-------------------|-------|----------------------------------------------------------------------------------|
| Has       | BOOKING — FLIGHT  | M : 1 | Many bookings can be associated with one flight; each booking is for one flight. |
| Allocated | BOOKING — SEAT    | M : 1 | Many bookings can reference seats over time; each booking references one seat.   |
| Operates  | AIRCRAFT — FLIGHT | 1 : M | One aircraft can operate many flights; each flight uses one aircraft.            |
| Contains  | AIRCRAFT — SEAT   | 1 : M | One aircraft contains many seats; each seat belongs to one aircraft.             |
| Departure | AIRPORT — FLIGHT  | 1 : M | One airport can be the departure airport for many flights.                       |
| Arrival   | AIRPORT — FLIGHT  | 1 : M | One airport can be the arrival airport for many flights.                         |

### 3. ER Diagram

The following diagrams show the entity-relationship design and the corresponding database table structure.



### Relational/table representation of the Airline Reservation System



**ER diagram showing entities, attributes, and relationships**

## 4. Relational Schema

The ER model is converted into the following relational schema:

```
PASSENGER(Passenger_ID PK, Name, Email, Passport_No, Phone)

BOOKING(Booking_ID PK, Booking_Date, Booking_Status, Passenger_ID FK, Flight_ID FK,
Seat_ID FK)

PAYMENT(Payment_ID PK, Amount, Payment_Date, Payment_Method, Payment_Status,
Booking_ID FK)

FLIGHT(Flight_ID PK, Flight_Number, Departure_Time, Arrival_Time, Status,
Aircraft_ID FK, Departure_Airport_ID FK, Arrival_Airport_ID FK)

AIRPORT(Airport_ID PK, Airport_Name, City, Country)

AIRCRAFT(Aircraft_ID PK, Model, Capacity, Manufacturer)

SEAT(Seat_ID PK, Seat_Number, Class, Aircraft_ID FK)
```

## 5. Primary Key and Foreign Key Identification

| Table     | Primary Key (PK) | Foreign Keys (FK)                                     |
|-----------|------------------|-------------------------------------------------------|
| PASSENGER | Passenger_ID     | —                                                     |
| BOOKING   | Booking_ID       | Passenger_ID, Flight_ID, Seat_ID                      |
| PAYMENT   | Payment_ID       | Booking_ID                                            |
| FLIGHT    | Flight_ID        | Aircraft_ID, Departure_Airport_ID, Arrival_Airport_ID |
| AIRPORT   | Airport_ID       | —                                                     |
| AIRCRAFT  | Aircraft_ID      | —                                                     |
| SEAT      | Seat_ID          | Aircraft_ID                                           |

## 6. Brief Relationship Explanation

**Passenger–Booking:** A passenger may make multiple bookings, while each booking is associated with one passenger. Therefore, the relationship is 1:M.

**Booking–Payment:** Each booking is associated with a payment record in this model. Therefore, it is represented as 1:1.

**Booking–Flight:** A flight can have many bookings, while each booking is for one flight. Therefore, the relationship is M:1 from Booking to Flight.

**Booking–Seat:** A booking references a seat, and the same seat can be referenced by bookings at different times. Therefore, it is represented as M:1 from Booking to Seat.

**Aircraft–Flight:** One aircraft can operate multiple flights, while each flight uses one aircraft. Therefore, the relationship is 1:M.

**Aircraft–Seat:** An aircraft contains multiple seats, while each seat belongs to one aircraft. Therefore, the relationship is 1:M.

**Airport–Flight (Departure):** An airport can be the departure point for many flights. Each flight has one departure airport, giving a 1:M relationship.

**Airport–Flight (Arrival):** An airport can be the arrival point for many flights. Each flight has one arrival airport, giving a 1:M relationship.

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